Laboratorio de Microeconomía I

Centro de Investigación y Docencia Económicas
Maestría en Economía - 2025
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Tarea 9

Exercises

Ex. 2.14 Consider the problem of insuring an asset against theft. The value of the asset is \$D, the insurance cost is \$I per year, and the probability of theft is p. List the four outcomes in the set A associated with this risky situation. Characterise the choice between insurance and no insurance as a choice between two gambles, each involving all four outcomes in A, where the gambles differ only in the probabilities assigned to each outcome.

Ex. 2.25 Consider the quadratic VNM utility function $U(w) = a + bw + cw^2$.

(a) What restrictions if any must be placed on parameters a, b, and c for this function to display risk aversion?

(b) Over what domain of wealth can a quadratic VNM utility function be defined?

(c) Given the gamble

$$g = ((1/2) \circ (w + h), (1/2) \circ (w - h)),$$

show that $CE < E(g)$ and that $P > 0$.

(d) Show that this function, satisfying the restrictions in part (a), *cannot* represent preferences that display *decreasing* absolute risk aversion.

Ex. 6.C.16^A An individual has Bernoulli utility function $u(\cdot)$ and initial wealth w. Let lottery L offer a payoff of G with probability p and a payoff of B with probability $1 - p$.

(a) If the individual owns the lottery, what is the minimum price he would sell it for?

(b) If he does not own it, what is the maximum price he would be willing to pay for it?

(c) Are buying and selling prices equal? Give an economic interpretation for your answer. Find conditions on the parameters of the problem under which buying and selling prices are equal.

(d) Let $G = 10$, $B = 5$, $w = 10$, and $u(x) = \sqrt{x}$. Compute the buying and selling prices for this lottery and this utility function.